

# Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 4

Clinical and Research devices often require wired connections, but wireless is taking over as security is

addressed.

ActiGraph.

Hardware Product Evolution



Sometimes the data are stored in a mixed format, text and low level binary

## Log Record Types

Note that some undocumented records are used for internal state or testing. They may be safely ignored.

| ID (dec) | ID (hex) | Type                           | Description                                                                                                                                                                                                                                              |
|----------|----------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0        | 0x00     | <a href="#">ACTIVITY</a>       | One second of raw activity samples packed into 12-bit values in XYZ order.                                                                                                                                                                               |
| 2        | 0x02     | BATTERY                        | Battery voltage in millivolts as a little-endian unsigned short (2 bytes).                                                                                                                                                                               |
| 3        | 0x03     | EVENT                          | Logging records used for internal debugging. These records notably contain information about idle sleep mode. When entering idle sleep mode, a record with payload 0x08 is created. When exiting idle sleep mode, a record with payload 0x09 is created. |
| 4        | 0x04     | HEART_RATE_BPM                 | Heart rate average beats per minute (BPM) as one byte unsigned integer.                                                                                                                                                                                  |
| 5        | 0x05     | LUX                            | Lux value as a little-endian unsigned short (2 bytes).                                                                                                                                                                                                   |
| 6        | 0x06     | <a href="#">METADATA</a>       | Arbitrary metadata content. The first record in every log is contains subject data in JSON format.                                                                                                                                                       |
| 7        | 0x07     | <a href="#">TAG</a>            | 13 Byte Serial, 1 Byte Tx Power, 1 Byte (signed) RSSI                                                                                                                                                                                                    |
| 9        | 0x09     | <a href="#">EPOCH</a>          | 60-second epoch data                                                                                                                                                                                                                                     |
| 11       | 0x0B     | <a href="#">HEART_RATE_ANT</a> | Heart Rate RR information from ANT+ sensor.                                                                                                                                                                                                              |
| 12       | 0x0C     | <a href="#">EPOCH2</a>         | 60-second epoch data                                                                                                                                                                                                                                     |
| 13       | 0x0D     | <a href="#">CAPSENSE</a>       | Capacitive sense data                                                                                                                                                                                                                                    |
| 14       | 0x0E     | <a href="#">HEART_RATE_BLE</a> | Bluetooth heart rate information (BPM and RR). This is a Bluetooth standard format.                                                                                                                                                                      |
| 15       | 0x0F     | <a href="#">EPOCH3</a>         | 60-second epoch data                                                                                                                                                                                                                                     |

Typically time based

- ▶ A header often defines parameters
- ▶ Each row is an entry
- ▶ Each column is a measure

# Typical Analysis

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Clinical Standards  
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Devices (PA)

Devices (Actigraphy)

Data

Typical Analysis

Analysis

Central Tendency

Measures of Variance

Measures of Variance

Measures of Variance

Standardized Measures

Comparison Tests

Analysis of Variance

Repeated Measures ANOVA

- ▶ Measures of central tendency
- ▶ Measures of variability
- ▶ Comparisons
- ▶ Trends (Binned, Change Point, Detrended Fluctuation)

# Central Tendency

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- ▶ Arithmetic Mean:

$$\bar{A} = \frac{1}{n} \sum_{i=1}^n a_i$$

- ▶ Geometric Mean<sup>1</sup>:

$$\bar{G} = \left( \prod_{i=1}^n x_i \right)^{\frac{1}{n}} = \sqrt[n]{x_1 x_2 \dots x_n}$$

- ▶ Harmonic Mean<sup>2</sup>:

$$\bar{H} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

- ▶ Median

- ▶ Mode

---

<sup>1</sup>useful whenever the quantities to be averaged combine multiplicatively

<sup>2</sup>most appropriate average for ratios and rates such as speeds, robust to outliers

## Measures of Variance

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## Clinical Standards of Wearable Devices (PA)

## Analysis

## Measures of Variance

- Range:

$$\text{range} = \max(X) - \min(X)$$

- Variance:

$$\text{Var}(X) = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

- Standard Deviation:

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$

- ▶ Pooled Standard Deviation:

$$\sigma_p = \sqrt{\frac{\sum_{i=1}^k (n_i - 1) \sigma_i^2}{\sum_{i=1}^k (n_i - 1)}} = \sqrt{\frac{(n_1 - 1) \sigma_1^2 + (n_2 - 1) \sigma_2^2 + \dots + (n_k - 1) \sigma_k^2}{n_1 + n_2 + \dots + n_k - k}}$$

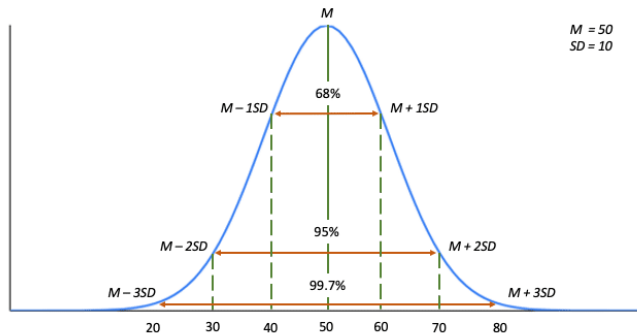
# Standard Deviation

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This is a “normal” curve, coming from theory driven by observations. One standard deviation is the inflection point on the curve.

Standard deviations in a normal distribution



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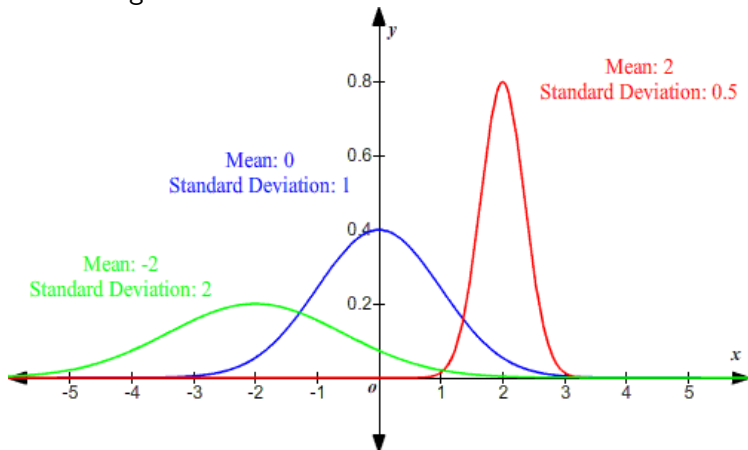
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# Pooled Standard Deviation

Suppose we have three participants and have measured the number of steps they take every day for a month. We have an average for each, and a standard deviation around those means for each participant. So then, how do we calculate the 'average' mean and standard deviation?





# Standardized Measures (Z-Score)

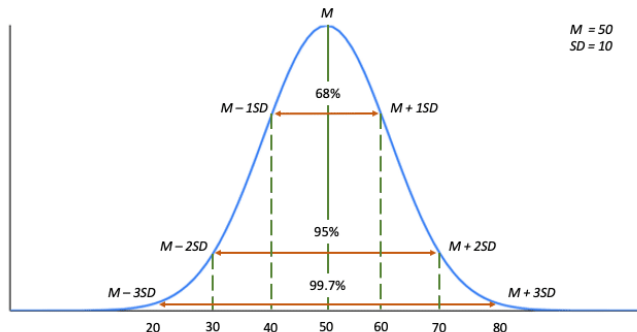
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What if we wanted to make some comparisons when measures are in different units? Or what what if the samples seem to have some bias, and we want to understand deviations from that bias?

$$Z_i = \frac{x_i - \mu}{\sigma}$$

Standard deviations in a normal distribution



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**Standardized Measures**

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# Comparison Tests

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**Comparison Tests**

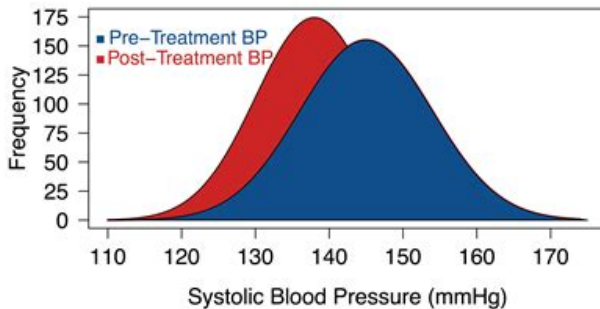
Analysis of Variance

Repeated Measures ANOVA

Suppose that we have samples of blood pressure before a treatment, and then after.

Likewise, what if we had two different groups we wanted to compare?

## Systolic Blood Pressure Before and After Treatment



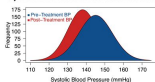
# Comparison Tests

Suppose that we have samples of blood pressure before a treatment, and then after.

Likewise, what if we had two different groups we wanted to compare?

We can use a t-test! This tests if the two samples are likely to have been sampled from the same population.

Systolic Blood Pressure Before and After Treatment



$$df = n_1 + n_2 - 2$$

$$\sigma_p = \sqrt{\frac{(n_1-1)\sigma_1^2 + (n_2-1)\sigma_2^2}{df}}$$

$$t = \frac{(\mu_1 - \mu_2)}{\sigma_p \cdot \sqrt{(\frac{1}{n_1}) + (\frac{1}{n_2})}}$$

$$cdf = tcdf(t, df)$$

| df  | Upper tail areas |       |        |        |        |
|-----|------------------|-------|--------|--------|--------|
|     | 0.1              | 0.05  | 0.025  | 0.01   | 0.005  |
| 1   | 3.078            | 6.314 | 12.706 | 30.821 | 63.687 |
| 2   | 1.886            | 2.920 | 4.303  | 6.965  | 9.525  |
| 3   | 1.638            | 2.353 | 3.182  | 4.541  | 5.841  |
| 4   | 1.533            | 2.132 | 2.776  | 3.747  | 4.604  |
| 5   | 1.476            | 2.015 | 2.571  | 3.365  | 4.032  |
| 6   | 1.440            | 1.943 | 2.447  | 3.143  | 3.707  |
| 7   | 1.415            | 1.895 | 2.365  | 2.998  | 3.499  |
| 8   | 1.397            | 1.860 | 2.306  | 2.896  | 3.355  |
| 9   | 1.385            | 1.833 | 2.262  | 2.821  | 3.250  |
| 10  | 1.377            | 1.812 | 2.228  | 2.764  | 3.183  |
| 11  | 1.369            | 1.796 | 2.201  | 2.718  | 3.136  |
| 12  | 1.366            | 1.782 | 2.179  | 2.681  | 3.099  |
| 13  | 1.360            | 1.771 | 2.160  | 2.650  | 3.062  |
| 14  | 1.356            | 1.761 | 2.145  | 2.624  | 3.037  |
| 15  | 1.354            | 1.753 | 2.131  | 2.602  | 3.017  |
| 16  | 1.351            | 1.746 | 2.118  | 2.583  | 2.998  |
| 17  | 1.349            | 1.740 | 2.107  | 2.567  | 2.983  |
| 18  | 1.348            | 1.734 | 2.100  | 2.552  | 2.970  |
| 19  | 1.346            | 1.729 | 2.093  | 2.539  | 2.959  |
| 20  | 1.345            | 1.725 | 2.088  | 2.528  | 2.949  |
| 21  | 1.343            | 1.721 | 2.084  | 2.518  | 2.941  |
| 22  | 1.342            | 1.717 | 2.079  | 2.509  | 2.934  |
| 23  | 1.341            | 1.714 | 2.076  | 2.500  | 2.928  |
| 24  | 1.340            | 1.711 | 2.074  | 2.493  | 2.923  |
| 25  | 1.339            | 1.708 | 2.070  | 2.485  | 2.917  |
| 26  | 1.338            | 1.706 | 2.066  | 2.479  | 2.912  |
| 27  | 1.337            | 1.703 | 2.062  | 2.473  | 2.907  |
| 28  | 1.336            | 1.701 | 2.060  | 2.467  | 2.903  |
| 29  | 1.335            | 1.699 | 2.058  | 2.462  | 2.900  |
| 30  | 1.334            | 1.697 | 2.056  | 2.457  | 2.896  |
| 40  | 1.303            | 1.664 | 2.023  | 2.423  | 2.794  |
| 45  | 1.296            | 1.657 | 2.000  | 2.390  | 2.660  |
| 120 | 1.289            | 1.650 | 1.980  | 2.350  | 2.617  |
| ∞   | 1.282            | 1.645 | 1.960  | 2.326  | 2.576  |

Adapted from Table B7 of R. A. Fisher and F. Yates, *Statistical Tables for Biological, Agricultural and Medical Research*, Sixth Edition, Pearson Education Limited, © 1963 R. A. Fisher and F. Yates.

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# Analysis of Variance (ANOVA)

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And what if we have three or more groups? A t-test isn't really going to work for us... why?

We can test if there exists a statistically significant difference among the groups first.

|          |    |    |    |
|----------|----|----|----|
| group    | 1  | 2  | 3  |
|          | 15 | 17 | 9  |
|          | 12 | 5  | 12 |
|          | 12 | 15 | 11 |
|          | 9  | 12 | 11 |
|          |    | 20 | 8  |
|          |    | 14 |    |
| n        | 5  | 7  | 6  |
| $\Sigma$ | 49 | 85 | 54 |

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_1 : H_0 \text{ is untrue}$$

$\alpha = 0.05$  (1/20 probability  
what we observe is from  
chance alone)

And what if we have three or more groups? A t-test isn't really going to work for us... why?

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|        |    |    |    |
|--------|----|----|----|
| group  | 1  | 2  | 3  |
|        | 15 | 17 | 9  |
|        | 12 | 5  | 12 |
|        | 12 | 15 | 11 |
|        | 9  | 12 | 11 |
|        |    | 20 | 8  |
|        |    | 14 |    |
| $n$    | 4  | 6  | 5  |
| $\sum$ | 48 | 83 | 51 |

$$N = \sum_{i=1}^k n_i = 15$$

Total Sum of Squares:

$$SS_T = \sum (X - \bar{X})^2$$
$$= \sum X^2 - \frac{(\sum X)^2}{N}$$

### Sum of Squares Between Groups:

$$\begin{aligned} SS_B &= \sum n_i (\bar{X}_i - \bar{X})^2 = \\ &= \sum_{i=1}^k \left( \frac{(\sum X_i)^2}{n_i} \right) - \frac{(\sum X)^2}{N} \\ &= \frac{48^2}{4} + \frac{83^2}{6} + \frac{51^2}{5} - \frac{182^2}{15} \end{aligned}$$

### Sum of Squares Within Groups:

$$SS_W = SS_T - SS_B$$

# Analysis of Variance (ANOVA)

And what if we have three or more groups? A t-test isn't really going to work for us... why?

We can test if there exists a statistically significant difference among the groups first.

MS is our mean squares

$m$  = number of groups

$N$  = number of observations

$n_j$  = number of observations in group  $j$

$x_{ij}$  = measure for observation  $i$  in group  $j$

$\bar{x}_j$  = mean for  $j$ th group

$\bar{X}$  = overall mean =  $\frac{\sum x_{ij}}{N}$

$$SS_{total} = \sum_{i=1}^N (x_{ij} - \bar{X})^2 = \sum_{i=1}^N x_{ij}^2 - \frac{(\sum_{i=1}^N x_{ij})^2}{N}$$

# Repeated Measures ANOVA

Suppose we have made measures at several time points on a group of subjects, such that we have something like this:

$n$  = number of subjects

$k$  = number of conditions

$x_{ij}$  = measure for subject  $i$  condition  $j$

$\bar{x}_j$  = mean for  $i$ th condition

$\bar{x}_i$  = mean for  $j$ th subject

$\bar{X}$  = overall mean

$$SS_{subjects} = \sum_{i=1}^n \sum_{j=1}^k (x_{ij} - \bar{x}_i)^2$$

$$SS_{conditions} = n \sum_{j=1}^k (\bar{x}_j - \bar{X})^2$$